

JYOTI

Senior Geospatial Analyst | GIS & Remote Sensing Specialist | Climate Intelligence & Spatial Data Analytics |
Urban Sustainability & Infrastructure Planning | Geospatial Automation & Mapping

[Email](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Results-driven Geospatial Analyst and GIS Specialist with over 6 years of experience in GIS, remote sensing, climate analytics, and spatial intelligence across urban sustainability, infrastructure planning, and environmental projects. Skilled in ArcGIS Pro, QGIS, Python, SQL, and Google Earth Engine for geospatial automation, spatial data management, QA/QC validation, thematic mapping, and decision-support analytics. Proven ability to deliver high-accuracy geospatial solutions, optimise GIS workflows, and transform complex spatial datasets into actionable insights for government, international, and global operational projects.

CORE EXPERTISE

- Urban Mapping & Environmental Analysis
- Geospatial Analytics & Spatial Intelligence
- Spatial Data Validation & QA/QC Compliance
- Climate Risk & Sustainability Assessment
- Infrastructure Planning and Site Suitability
- GIS Automation & Workflow Optimisation
- Technical Reporting & Cartography
- Remote Sensing & Earth Observation

TECHNICAL SKILLS

- GIS & Spatial Tools: QGIS, ArcGIS Pro, SNAP, ERDAS Imagine
- Programming & Automation: Python (geopandas, numpy), SQL, Google Earth Engine (GEE)

WORK EXPERIENCE

Senior Program Associate / Program Associate

WORLD RESOURCES INSTITUTE, INDIA (FULL-TIME)

September 2021 – Present

- Spearheaded geospatial and climate analytics initiatives across 5+ large-scale state and city-level sustainability projects, ensuring all GIS outputs aligned with approved technical standards, methodologies, and spatial compliance frameworks.
- Produced high-precision thematic maps, technical visualisations, and analytical reports using QGIS and ArcGIS Pro, translating complex spatial datasets into actionable insights supporting climate policy frameworks across 6 Indian cities.
- Led multi-stage project execution for urban sustainability and climate resilience initiatives while mentoring junior associates and interns, ensuring timely project delivery and adherence to GIS methodologies and reporting standards.
- Conducted large-scale spatial data validation, QA/QC checks, and integrity assessments for government and institutional stakeholders, including NITI Aayog, ensuring high-accuracy GIS deliverables and data reliability.
- Developed and automated geospatial workflows using Python and Google Earth Engine, significantly improving efficiency in spatial data retrieval, time-series analysis, and environmental monitoring processes.

GIS Analyst

X, ALPHABET (PART-TIME CONTRACT)

July 2021 – March 2025

- Managed and monitored global spatial datasets while maintaining data accuracy, geospatial consistency, and adherence to established GIS methodologies across operational projects.
- Leveraged advanced remote sensing techniques and geospatial analysis tools to generate high-precision insights, improving project turnaround efficiency and operational decision-making.

- Supported GIS system integration initiatives to identify optimal global site locations through spatial intelligence, remote sensing analysis, and location-based analytics.
- Collaborated with cross-functional global teams to streamline GIS data workflows, improve reporting accuracy, and support strategic infrastructure planning initiatives.

Climate Analytics Consultant (Contract)

INTERNATIONAL CENTER FOR AGRICULTURAL RESEARCH IN DRY AREAS (ICARDA) March 2021 – June 2021

- Developed a custom Google Earth Engine application for visualisation and export of high-volume CMIP5 NEX-GDDP climate projection datasets, supporting long-term climate mitigation and environmental planning strategies.
- Enabled seamless processing, interpretation, and accessibility of large-scale climate datasets for sustainability-focused decision-making in arid and environmentally vulnerable regions.

Consultant

WORLD RESOURCES INSTITUTE (WRI) INDIA August 2020 – August 2021

- Delivered geospatial analysis and remote sensing support across multiple environmental and urban development projects involving spatial analysis, mapping, and technical GIS reporting.
- Contributed to environmental research initiatives through advanced spatial data interpretation, cartographic visualisation, and GIS-based analytical support.
- Assisted project teams in improving spatial data workflows, map production quality, and geospatial reporting efficiency across ongoing initiatives.

Intern

MOODY'S (FORMERLY RISK MANAGEMENT SOLUTIONS) INDIA July 2019 – March 2020

- Collaborated on semi-automated asset extraction frameworks that improved spatial data accuracy and operational efficiency by 30% across GIS workflows and technical processes.
- Built internal GIS support tools to streamline spatial data queries, optimise technical operations, and enhance the utilisation of GIS software across teams.
- Assisted in geospatial workflow optimisation, technical analysis, and spatial data processing for risk and infrastructure-related projects.

EDUCATION AND FELLOWSHIPS

Terra.do (Online) Oct 2025 – Jan 2026
Learning for Action: Climate Change

National Institute of Technology, Karnataka, India 2018 – 2020
Master of Technology in Remote Sensing and GIS (CGPA 9.43)

Institute of Engineering and Technology, Lucknow, Uttar Pradesh, India 2014 – 2018
Bachelor of Technology in Civil Engineering (Percentage – 77.7)

LICENSES AND CERTIFICATIONS

- Certificate of Completion for PyQGIS Masterclass – **SpatialThoughts**
- Certificate of Completion for Advanced QGIS – **SpatialThoughts**
- Certificate of Appreciation, Best Thesis Awards (Kerala Chapter) 2020 – **IEEE-GRSS**
- Health GIS: Geoinformatics for COVID-19 – **IIRS**
- Machine Learning with TensorFlow on Google Cloud Platform Specialization – **Coursera**
- SAR for Landcover Applications – **NASA**
- SAR for Disasters and Hydrological Applications – **NASA**